Time Series Analysis

PARAMI UNIVERSITY

Time Series Analysis

Today's Topic: Seasonal Autoregressive Integrated Moving Average Model

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The Box-Jenkins Method

- If a time series is nonstationary, we make it stationary by differencing.
- Then use an appropriate ARMA model to fit the differenced data.
- This is known as the Box-Jenkins modeling approach of a step-by-step procedure.
- Identify -> Estimate -> Diagnose -> Forecast

Seasonal Differencing

- A nonstationary time series $\{X_t\}$ with trend and seasonal period s.
- The differenced series is given by

$$Y_t = \nabla^d \nabla_s^D X_t = (1 - B)^d (1 - B^s)^D X_t$$

  then we get the stationary series.
- Where, B is the backshift operator: $\nabla^d = (1 - B)^d$ is the ordinary differencing of order d; and $\nabla_s^D = (1 - B^s)^D$ is the seasonal differencing of order D.
- In most case $D \leq 2$.
- After differencing, $d + Ds$ data points will drop and Y_t is fewer data points than X_t.

Seasonal Differencing

- The backshift operator is denoted by B and defined as

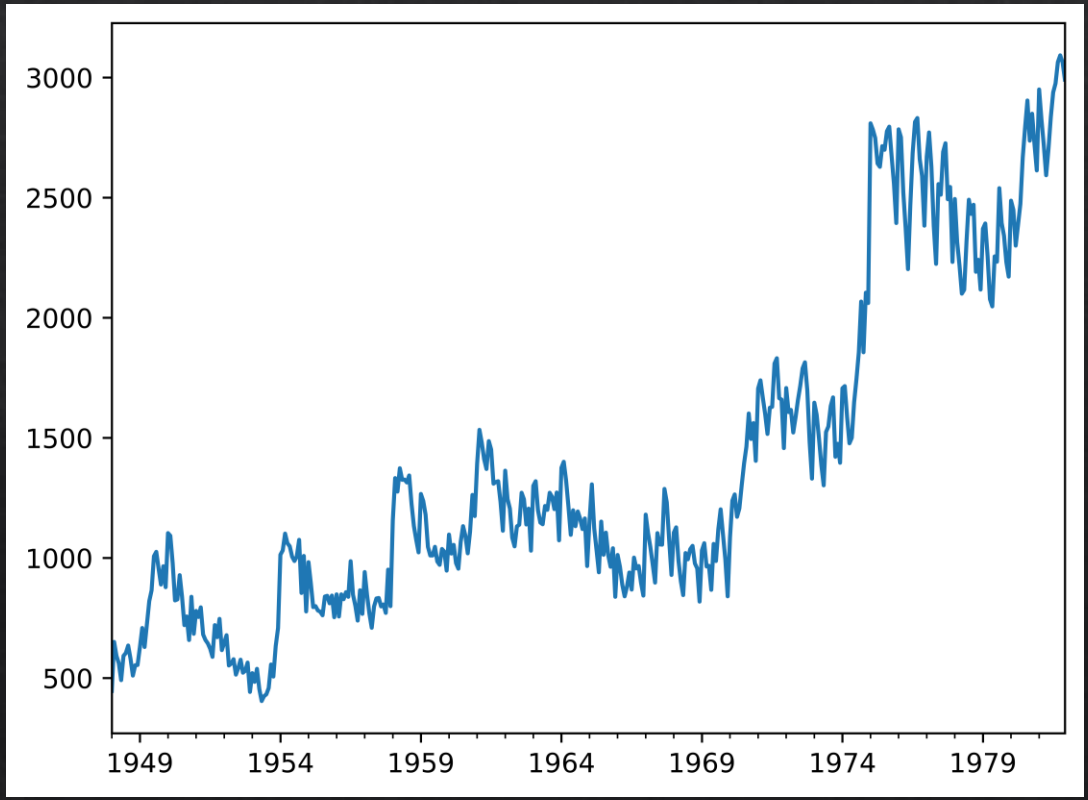
$$By_t = y_{t-1}$$

- $B^1 y_t = y_{t-1}$
- $B^2 y_t = y_{t-2}$
- $B^s y_t = y_{t-s}$

Make it Stationary

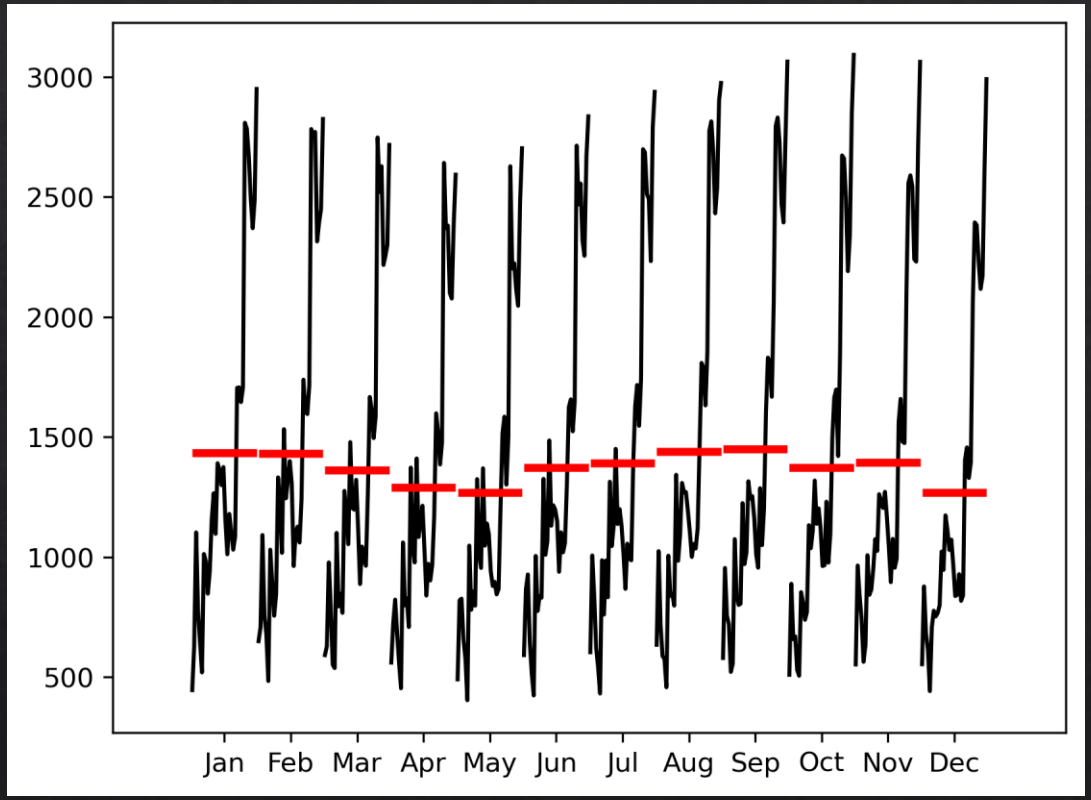
America Monthly Employment Figures for Females Aged 20 Years and Over

- We will use the US monthly employment figures for females aged 20 years and over from 1948 to 1981.



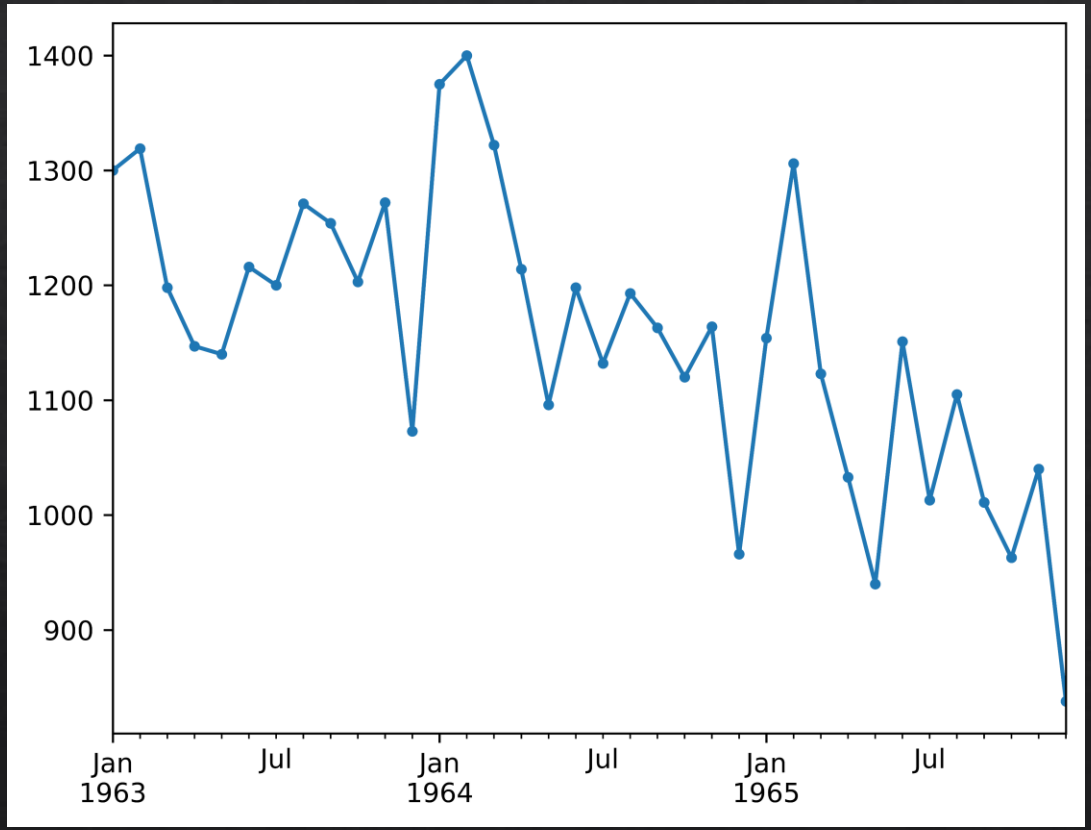
Make it Stationary

Seasonal Plots of females aged 20 and over



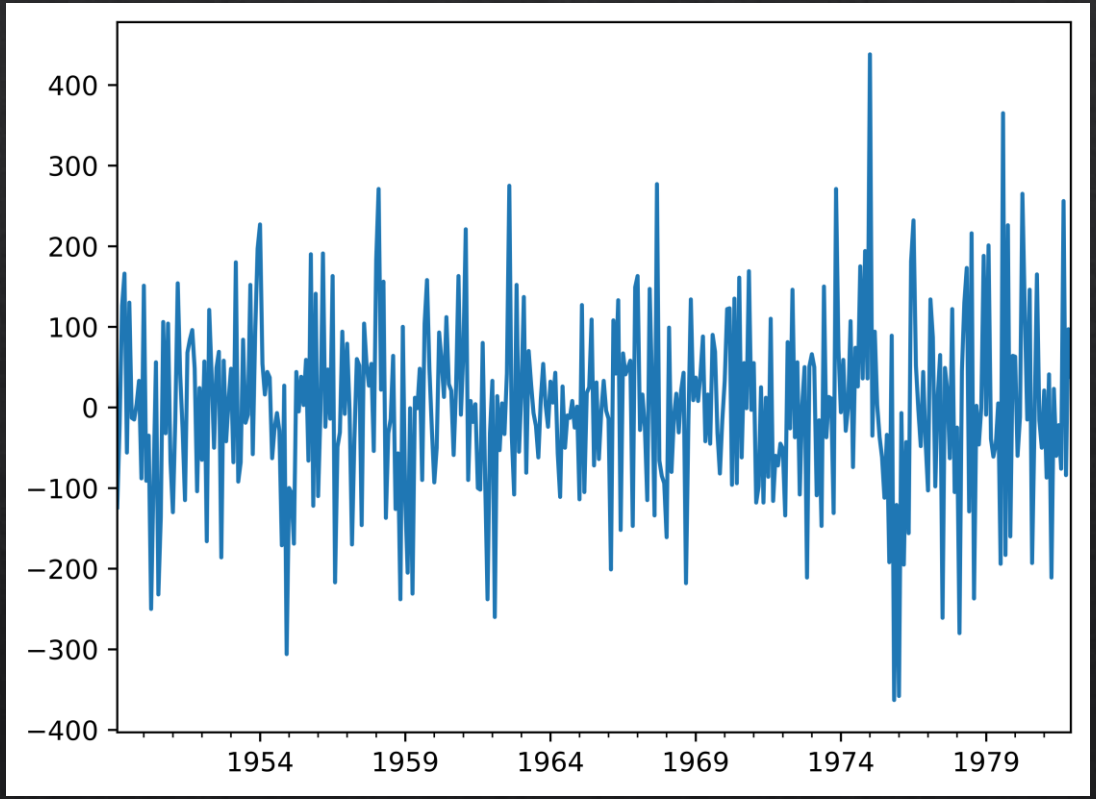
Make it Stationary

Time Series Plot of the subseries (1963-1965)



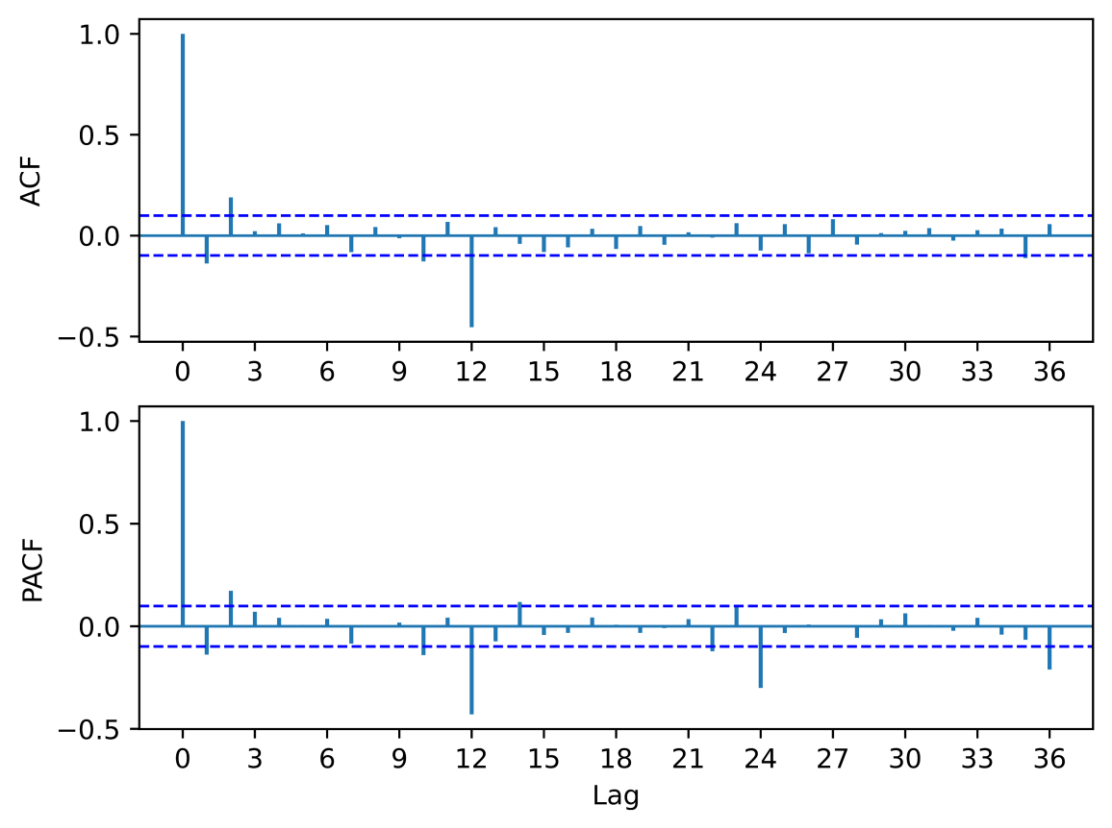
Make it Stationary

Time Series Plot after differenced data



Make it Stationary

ACF and PACF



SARIMA Models

- If d and D are nonnegative integers and B is the backshift operator, then time series $\{X_t\}$ is a SARIMA(p, d, q)(P, D, Q)$_s$ process with seasonal periods s.
- If the differenced series $Y_t = (1-B)^d(1-B^s)^D X_t$ is stationary, and $\{X_t\}$ satisfies the following model

$$\varphi(B)\Phi(B^s)(1-B)^d(1-B^s)^D X_t = \theta(B)\Theta(B^s)\varepsilon_t.$$

- If is stationary, $d = D = 0$, then the model is reduced to

$$\varphi(B)\Phi(B^s)X_t = \theta(B)\Theta(B^s)\varepsilon_t$$

  which is SARMA(p, q)(P, Q)$_s$ model.
- If p=q=d=0, then the model is reduced to

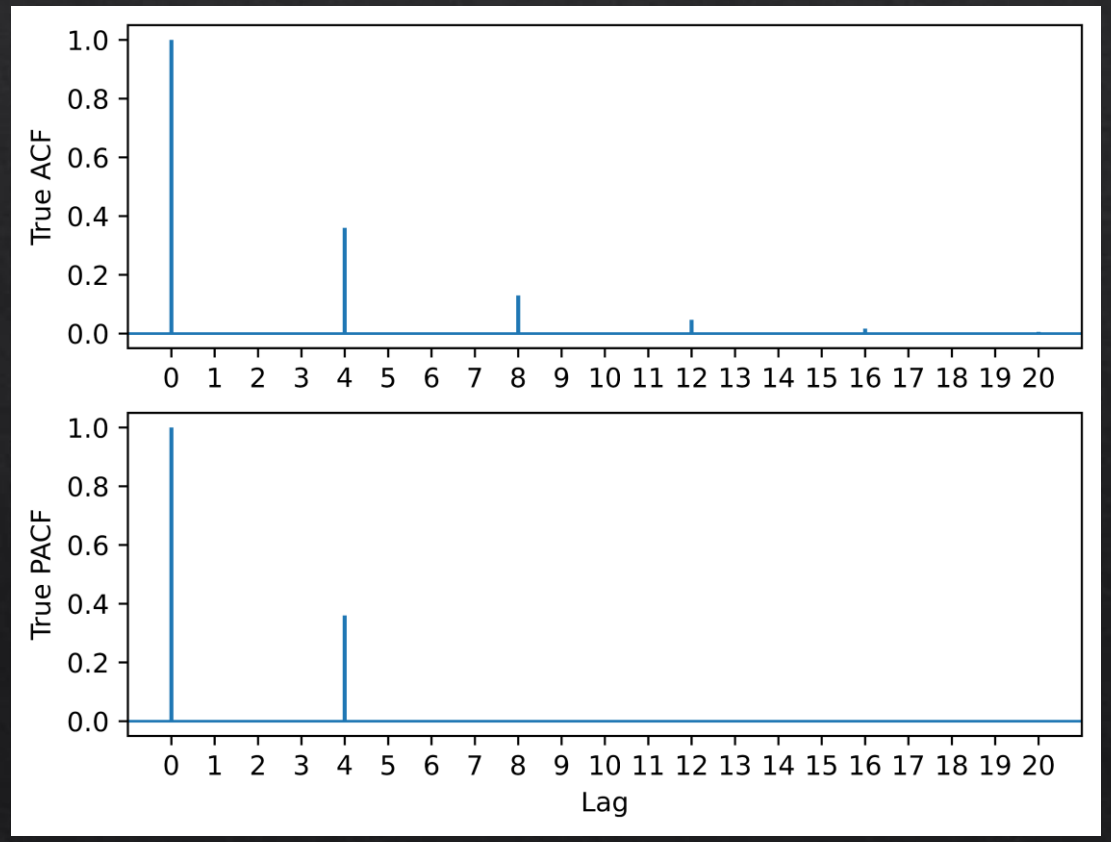
$$\Phi(B^s)(1-B^s)^D X_t = \Theta(B^s)\varepsilon_t$$

  which is pure SARIMA(P, D, Q)$_s$ and if D=0, which is pure SARMA(P,Q)$_s$.
- If Q=0, then pure SAR(P)$_s$ and if P=0 then pure SMA(Q)$_s$.

Pure SAR(P)$_s$

Pure SAR(1)$_4$

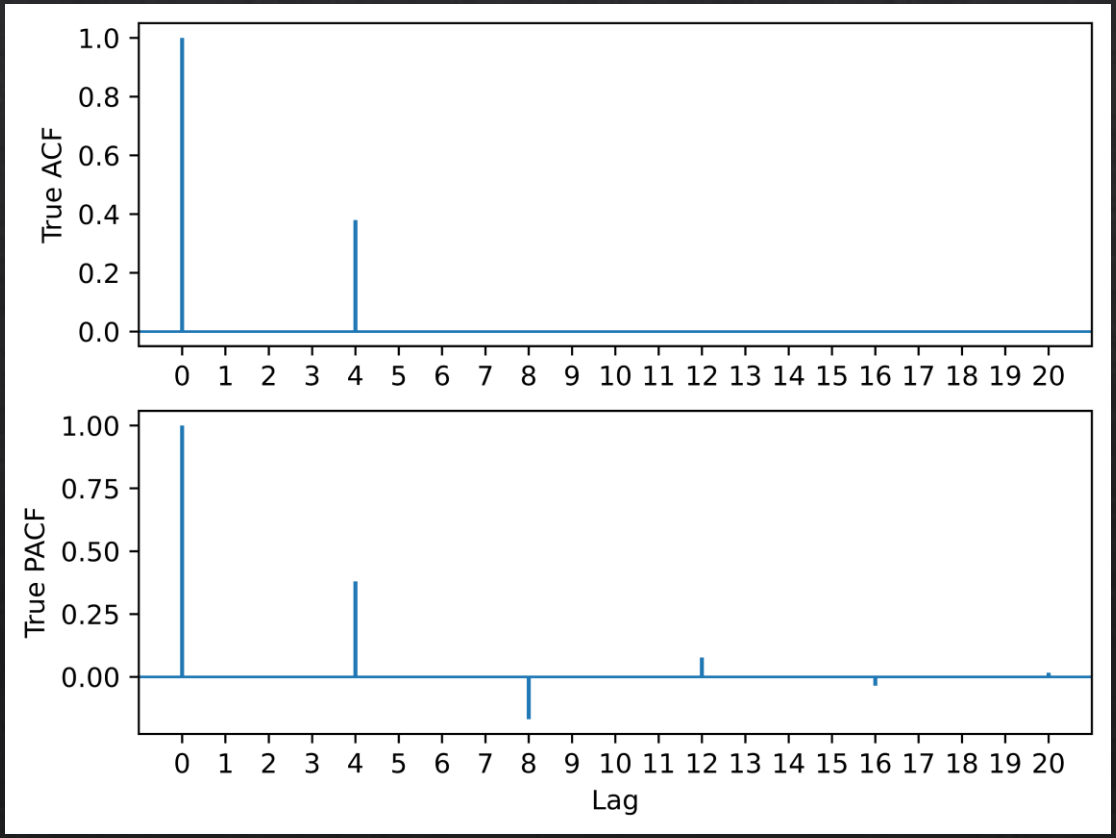
$(1 - 0.36B^4)X_t = \varepsilon_t$ Or $X_t = 0.36X_{t-4} + \varepsilon_t$



Pure SMA$(Q)_s$

Pure SMA$(1)_4$

$X_t = (1 - 0.46B^4)\varepsilon_t$ Or $X_t = \varepsilon_t + 0.46\varepsilon_{t-4}$

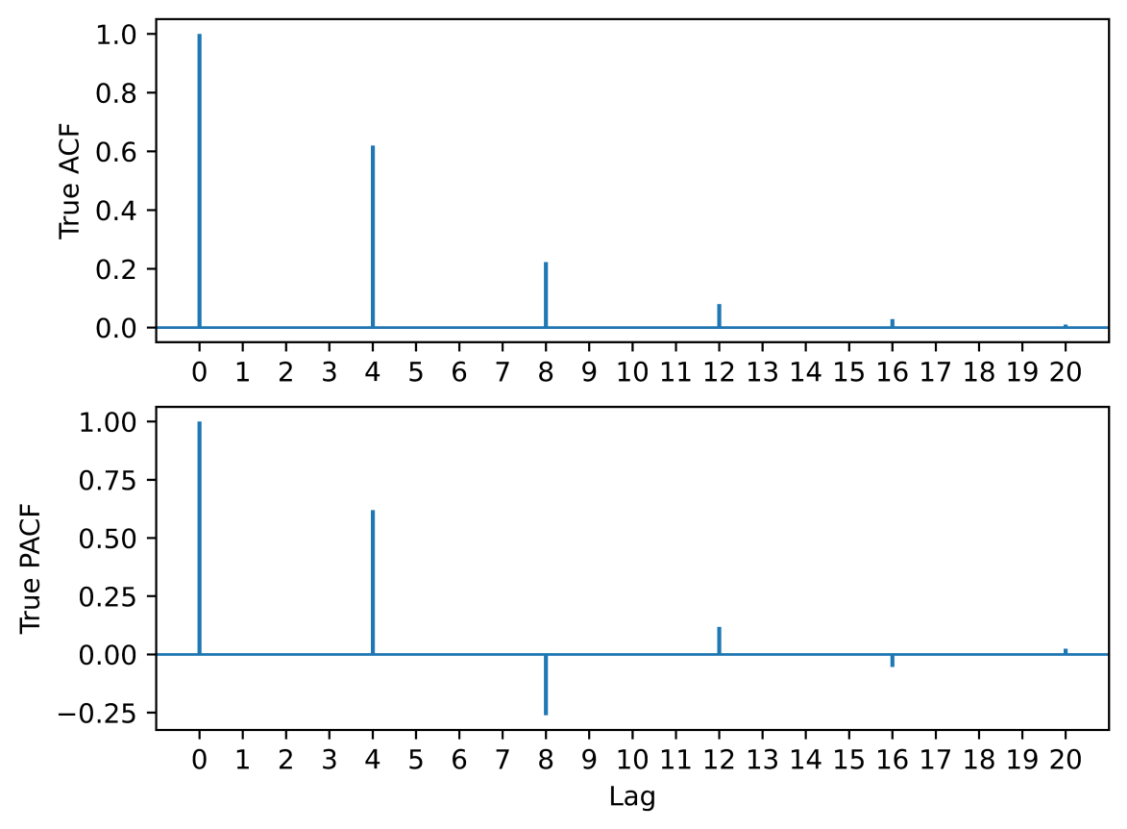


Pure SARMA(P, Q)$_s$

Pure SARMA(1, 1)$_4$

$(1 - 0.36B^4)X_t = (1 - 0.46B^4)\varepsilon_t$ Or

$X_t - 0.36X_{t-4} = \varepsilon_t + 0.46\varepsilon_{t-4}$



SARMA models

Table 1. Properties of the ACF and PACF.

	SMA$(Q)_s$	SAR$(P)_s$	SARMA$(P, Q)_s$ $(P > 0, Q > 0)$
ACF	Cuts off after lag Qs	Tails off at lag ks $(k = 1, 2, \cdots)$	Tails off at lag ks $(k = 1, 2, \cdots)$
PACF	Tails off at lag ks $(k = 1, 2, \cdots)$	Cuts off after lag Ps	Tails off at lag ks $(k = 1, 2, \cdots)$

Note: whether the ACF or PACF, their values are all zero at lag $l \neq ks$ for any integer k.

Conclusion

- There is no strict rule for choosing d and D, so determine the values by examining ACF and PACF plots, and use the stationarity test, like KPSS, and d and D are usually 0 or 1 and rarely exceed 2.
- Use the same ACF and PACF plots for examining p, q, and P and Q, but determine p and q from small lags and P and Q from seasonal lags.
- In some cases, the model order cannot be clearly identified from ACF and PACF plots. Therefore, we consider several candidate models and select the best one using information criteria such as AIC, BIC, or HQIC.